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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 2: Introduction to Transportation Systems-II

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Dept. of Civil Engineering



Contents

- Transportation Systems
- Transport Modes
- Factors Influencing Efficiency of a Transport System
- Factors Influencing Development of a Transportation System
- Transportation System Components
- Multimodal Transport



Transportation System ?

- A *transportation system* can be defined as

"the combination of elements and their interactions, which produce the demand for travel within a given area and the supply of transportation services to satisfy this demand."

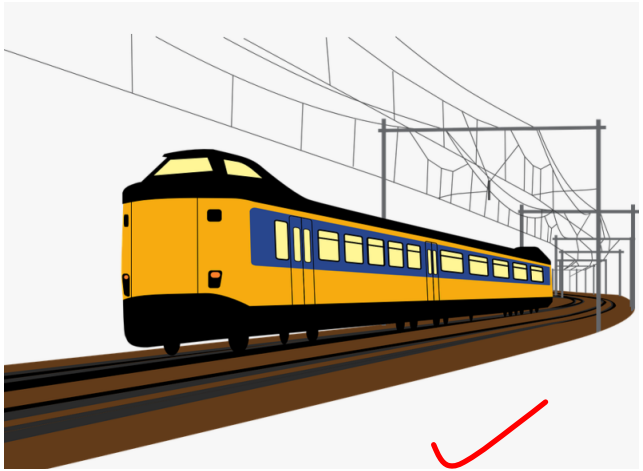


- Transport modes
  - The various networks used to transport people and/or goods.
    - Includes Motorized and Non Motorized Transport (NMT) modes

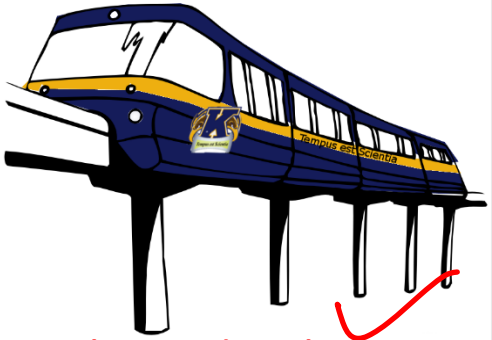
Source: (Ennio Cascetta, 2001)

Transport Modes

Railways



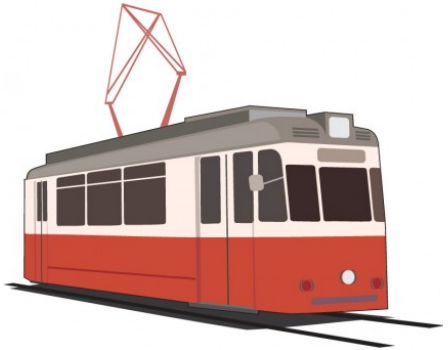
Surface Railways



Elevated Railways



Underground (Tube) Railways

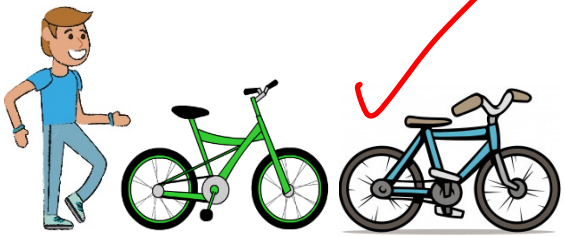


Light Rail Transit (Trams)

Source: (L. R. Kadiyali 2017)

Transport Modes

Road Transport



Non Motorized modes (Walking & Bicycles)



Motorcycles



Public Transport



Private Cars



Heavy Vehicles

Source: (L. R. Kadiyali 2017)

Transport Modes

Air Transport



Commercial Aircrafts



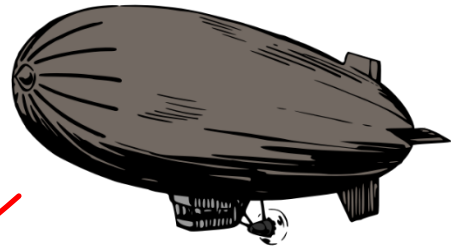
Helicopters



Hot Air Balloons



Private Jets



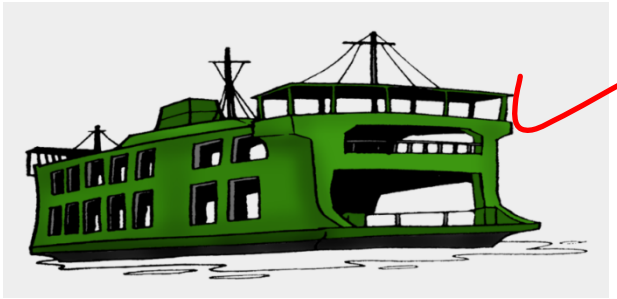
Airships or Blimps

Source: (L. R. Kadiyali 2017)

Transport Modes

Water Transport

- Inland Water Transport
- International Water Transport



Ferry boats



Small Boats



Cruise ships



Cargo ships



Submarines

Source: (L. R. Kadiyali 2017)

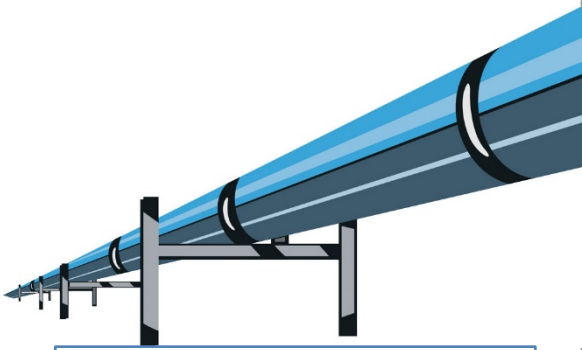
Transport Modes

Pipelines



Pipelines carrying water from a water reservoir to the target destination

Source: (https://www.worldatlas.com/articles/top-20-countries-by-length-of-pipeline.html)



Source: (L. R. Kadiyali 2017)

Transport Modes

Ropeways



Monocable Ropeway



Bicable Ropeway



Tricable Ropeway

Source: (Alshalalfah, et. al 2012)

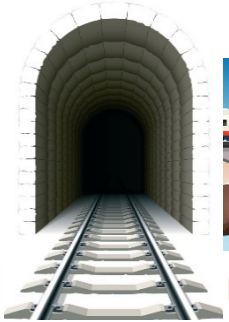
Transportation Infrastructure

Railway infrastructure

- Infrastructure for all types of railways including subways and elevated cable cars
  - Includes railway lines and other structures, buildings and equipments, together with the land located on railway premises.

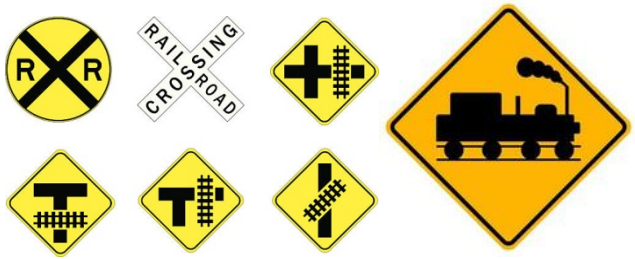


Railway tracks

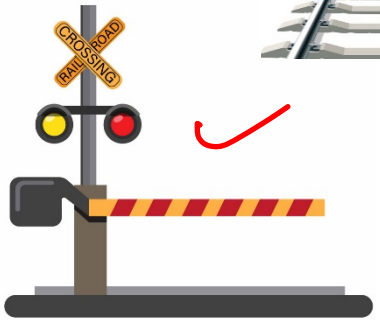




Railway Tunnels and Bridges



Railway Signages



Railway Crossings

Source: (The Railway technical website, http://www.railway-technical.com/infrastructure/)

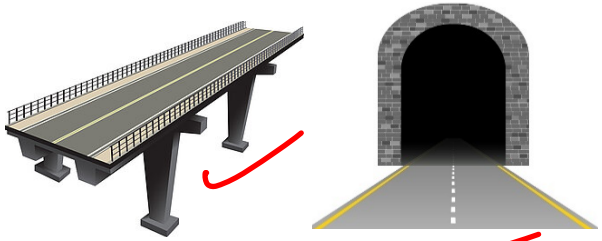
Transportation Infrastructure

Road infrastructure

- Transport Infrastructure includes
  - Paved roads and unpaved roads and all associated physical assets such as signages, traffic lights, street lights, pedestrian walkways, all earth works such as speed brakers and mile stones, drainage works and structures such as culverts, bridges and buildings



Paved and Unpaved roads



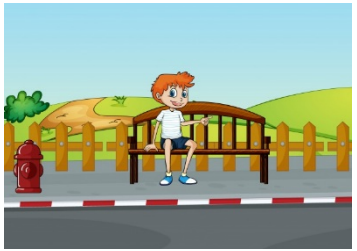
Bridges and Tunnels



Traffic Signages



Traffic lights and Street lights



Pedestrian facilities

Source: (Roberts, 2004)

Transportation Infrastructure

Waterways

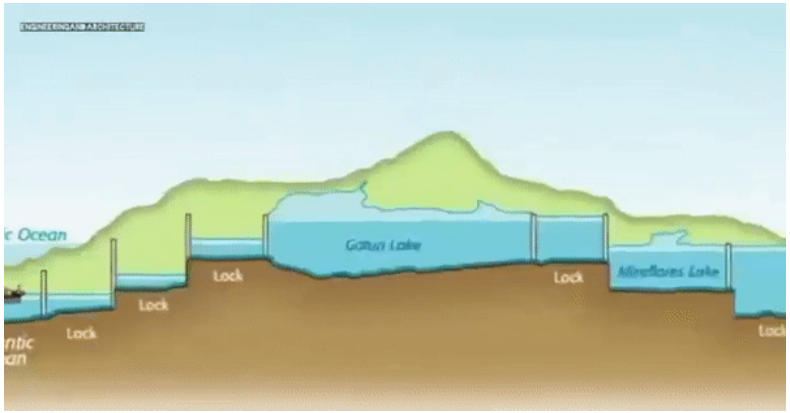
- Includes infrastructure for navigable waterways such as Canals and rivers
  - Navigation Buoys (Inland waterways)
  - Light houses
  - Navigable aqueducts
  - Canal tunnels



Navigation Buoys



Light house



Working of Panama Canal

Source: (https://i.redd.it/s4nd0hrj3eh51.gif)



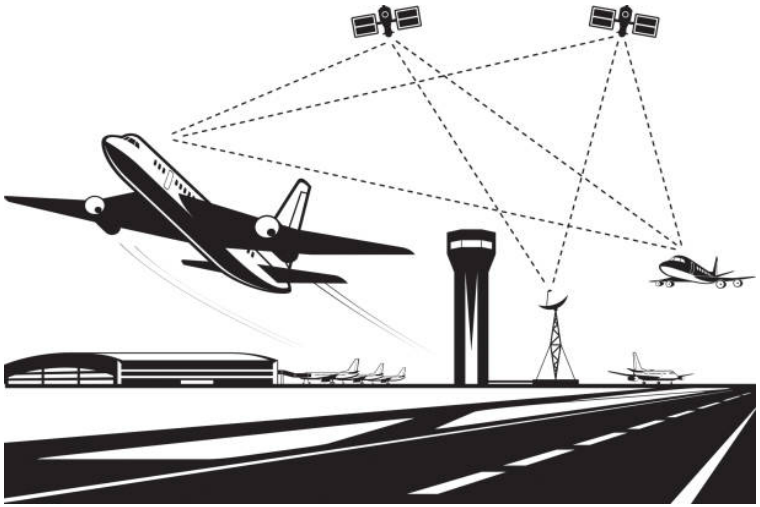
Navigable Aqueducts

Source: (https://www.homedit.com/the-longest-navigable-aqueduct-in-the-world/)

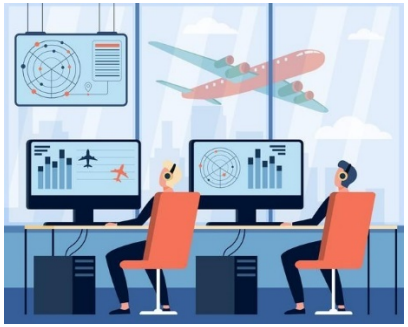
Transportation Infrastructure

Air Traffic Control

- Infrastructure for Air traffic such as
  - Air routes and traffic control facilities
  - Air navigation systems
  - Air traffic flow management facilities



Air traffic control tower



Control room



Aircraft hanger



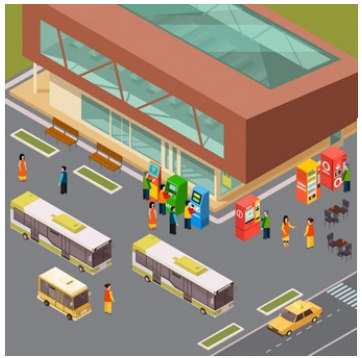
Apron

Source: (International Air Transport Association (IATA), https://www.iata.org/en/youandiata/airports/)

Transportation Infrastructure

Terminals and Stations

- Includes terminals and waiting stations for passenger boarding and arrival for buses, railways, waterways and airways
- Passenger comfort, convenience and service is of prime importance.





Bus terminal



Railway Station



Airport terminal

Transportation Infrastructure

Ports/ Seaports

- Docking facilities and facilities for transfer of people and goods



Docking facilities



Container loading



Port Cranes

Source: (Port Infrastructure, Waterborne, https://www.waterborne.eu/)

Transportation Infrastructure

Cycling Infrastructure

- Infrastructure facilities such as bike paths, parking facilities



Bike paths

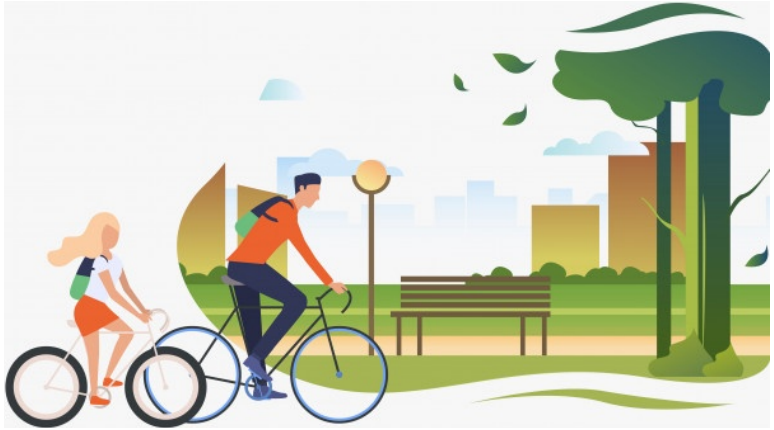


Parking facilities

Transportation Infrastructure

Living streets

- Living streets designed for multiple uses, restricting speed of vehicles, pedestrian importance, incorporating children, recreation and green spaces.





Factors influencing Efficiency of a Transportation System ?

11 Factors

- Speed
- Safety
- Adequacy
- Frequency
- Regularity
- Integration
- Responsibilities
- Comfort
- Cheapness
- Fuel efficiency
- Employment Generation



Source: (L. R. Kadiyali 2017)

Factors influencing Efficiency of a Transportation System ?

- Speed
  - Essence of transport
  - Faster transport, Reduces overall cost, reduces storage and inventorying
  - Faster turn-around

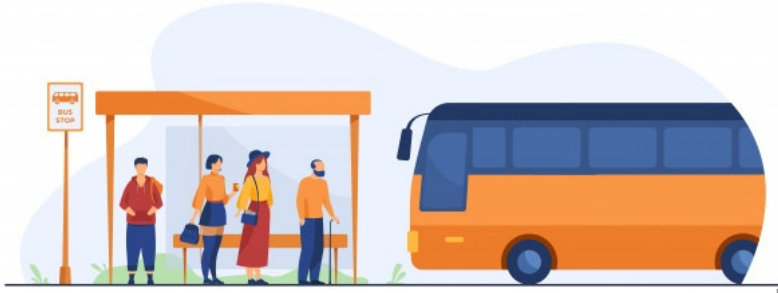


- Safety
  - Safety of passengers and goods
  - Highest possible standard of construction, maintenance and operation
  - Protection against theft, fire and damage in case of goods

Source: (L. R. Kadiyali 2017)

Factors influencing Efficiency of a Transportation System ?

- Adequacy
  - Represents the capacity of a transport mode
  - Should have sufficient capacity

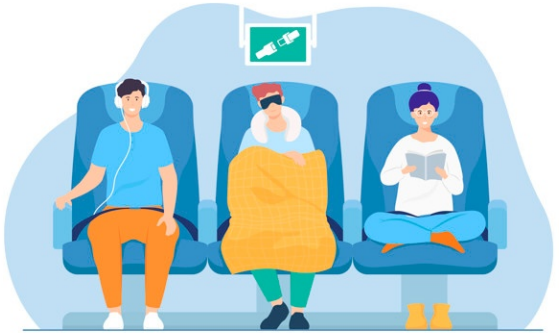


- Frequency
  - Important characteristic of a transport mode
  - Related to capacity consolidation and storage
  - Assured frequency attracts people and goods
- Regularity
  - Punctuality and the ability to adhere to the scheduled service
  - Regular service gives confidence to its clients

Source: (L. R. Kadiyali 2017)

Factors influencing Efficiency of a Transportation System ?

- Integration
  - Where the mode requires transfer from one mode to another
  - A system with few transfers is considered an efficiently integrated system
- Responsibility
  - Ability to meet liabilities for loss or damage
- Comfort
  - Essential factor for passenger transport



Source: (L. R. Kadiyali 2017)

Factors influencing Efficiency of a Transportation System ?

- Cheapness
  - Main consideration for consumer preference
  - Depends on total facility resources cost including construction, maintenance and operational costs
- Fuel efficiency
  - The Global energy crisis demands consumers to adopt fuel efficient transport modes
- Employment Generation
  - Any mode of transport having good employment potential should be favoured



Source: (L. R. Kadiyali 2017)

Factors influencing Development of a Transportation System

Factors

Local Scale



Regional Scale



National/ Transnational Scale



Global Scale



- Environmental factors
- Historical factors
- Technological factors
- Political factors
- Economic factors

Source: (Jean-Paul Rodrigue, 2020)

Factors influencing Development of a Transportation System

Local Scale

- Environmental
  - Hydrography and Geomorphology
- Historical
  - Cultural and Settlement patters
- Technological
  - Roads
- Political
  - Zoning
- Economic
  - Employment and distribution

Source: (Jean-Paul Rodrigue, 2020)

Factors influencing Development of a Transportation System

Regional Scale



- Environmental
  - Climate
- Historical
  - Urban System
- Technological
  - Railways and Canals
- Political
  - Taxation and Regulations
- Economic
  - Modal competition and complementarity

Source: (Jean-Paul Rodrigue, 2020)

Factors influencing Development of a Transportation System

National/ Transnational Scale



- Environmental
  - Distance
- Historical
  - Nation state/ Colonialism/ Imperialism
- Technological
  - Corridors and Sea routes
- Political
  - Trade agreements
- Economic
  - Markets

Source: (Jean-Paul Rodrigue, 2020)

Factors influencing Development of a Transportation System

Global Scale

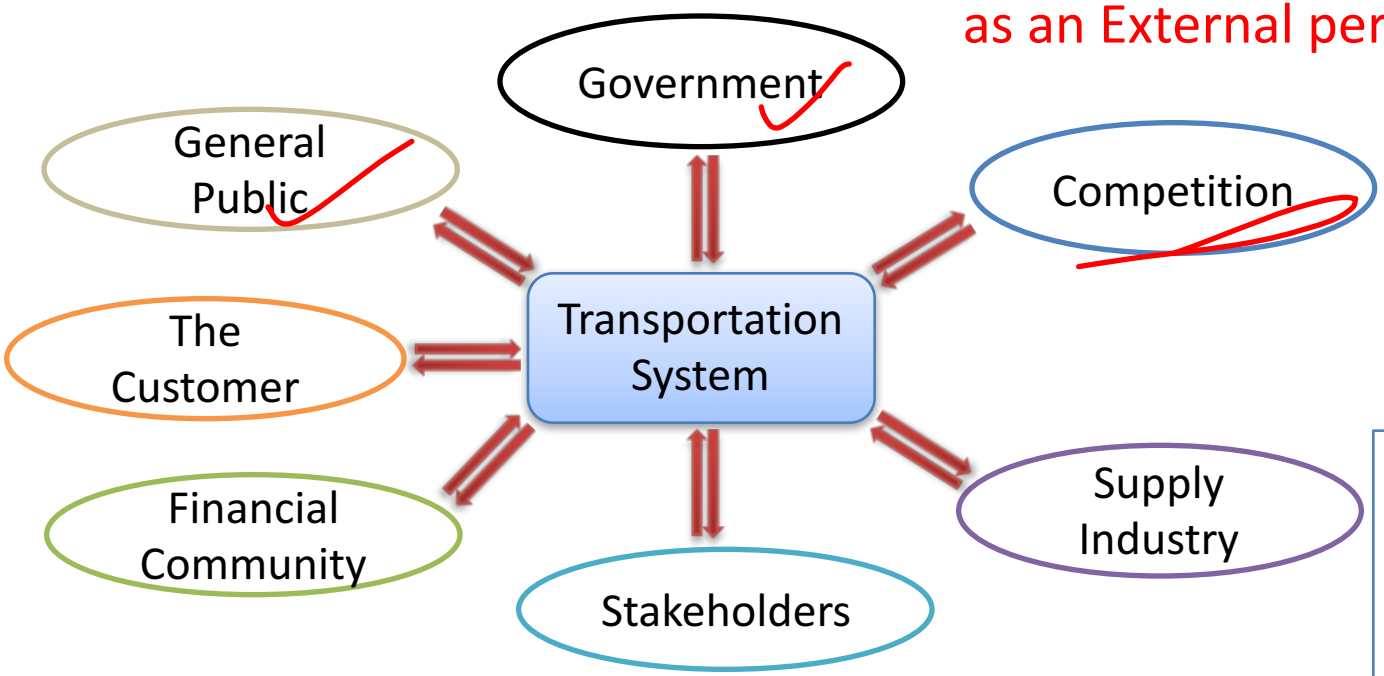


- Environmental
  - Oceanic masses
- Historical
  - Globalization
- Technological
  - Air transport and Telecommunications
- Political
  - Multilateral agreements (WTO)
- Economic
  - Interdependency and comparative advantages

Source: (Jean-Paul Rodrigue, 2020)

Transportation System Components ?

- Transportation System Components as an External perspective



Source: (Joseph M. Sussman, 2006)

Transportation System Components

Government

- Taxes
- Safety Regulations
- User Fees
- Financial Regulation
- Entry and Exit from Market
- Monopolies
- Provider of Infrastructure
- Provider of Funding





Source: (Joseph M. Sussman, 2006)

Transportation System Components

Competition

- Intra-modal
- Inter-modal (Rail vs Truck)
- For the Consumer's money
  - Buy a two wheeler or take a vacation
- Transportation/Communication System Competition

Common error: Viewing the Competition as Static



Source: (Joseph M. Sussman, 2006)

Transportation System Components

Supply Industry

- Vehicle Providers
- Infrastructure Builders
  - Low-tech
  - High-tech
- Components
  - Electronics
  - Materials
- Research Community
- Insurance Industry

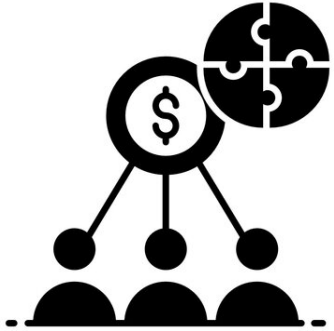


Source: (Joseph M. Sussman, 2006)

Transportation System Components

Stakeholders

- People/Organizations who are not customers or suppliers but are nonetheless concerned
  - Environmental Community
  - Abutters
  - The Public- Concerned with quality of life, national defense, economic development, as enabled by the transportation system



Financial Community

- Provider of:
  - Equity
  - Debt

Source: (Joseph M. Sussman, 2006)

Multimodal Transport



- Combining multiple modes of transportation to transport people/goods effectively from point of source to point of destination.

- Also called as "Combined transport" or "Intermodal transport"

- Ensures seamless travel by:
  - Vehicle mode Integration (Car, BRTS, Metro etc.)
  - Service Integration (Private & Public modes)
  - Infrastructure Integration (Bus & metro stations, Parking facilities etc.)

Multimodal Transport Concept



Access Modes

Walking
Bicycle,
Rickshaws
2 wheeler
Bus

BRTS/ MRTS

Transfer stations

Egress Modes

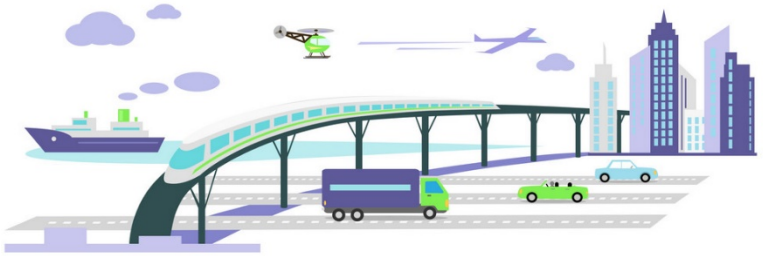
Walking
Bicycle,
Rickshaws
2 wheeler
Bus

Multimodal Transport

Challenges



- Poor Infrastructure facilities
- Intermodal linkage at regional and local levels



- Current aged transport modes
- Enhance Information and Communication technologies
- Modernization of public passenger terminals
- Role of Government to facilitate MMT

Conclusions

- Transportation Systems includes all modes of transport (road, rail, air, waterway, pipeline and cable) along with its infrastructure facilities.

- An efficient Transportation System comes with efficient policy measures and good infrastructure facilities, integrated modes of transport and integration with ICT facilities to tackle congestion, pollution and other negative externalities affecting environment.

- Sustainable solutions and measures are the future of Transportation systems.

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Thank You